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(54) **REMOTE POWER SYSTEM RACK AND ASSEMBLY**

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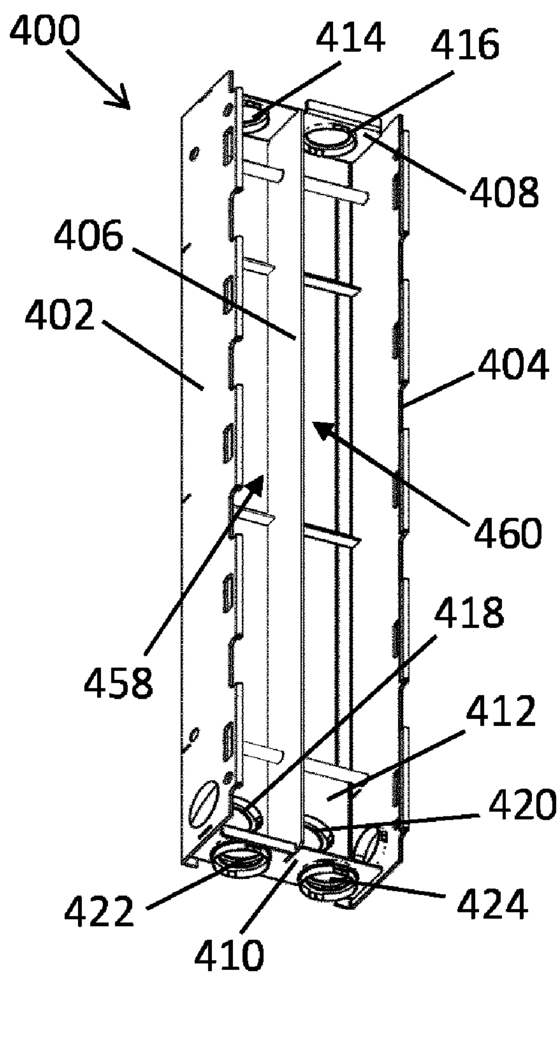
CPC ***H02G 3/0456*** (2013.01); ***H02G 3/0418*** (2013.01); ***H02G 3/263*** (2013.01)

(57) **ABSTRACT**

A mounting rack for attachment of a remote power unit to a structure includes a first side frame, a second side frame, and a separator plate. The first side frame and the separator plate provide an alternating current (AC) wiring pathway therebetween for routing a first electrical cable. The second side frame and the separator plate provide a direct current (DC) wiring pathway therebetween for routing a second electrical cable. The AC wiring pathway and the DC wiring pathway are separated by the separator plate.



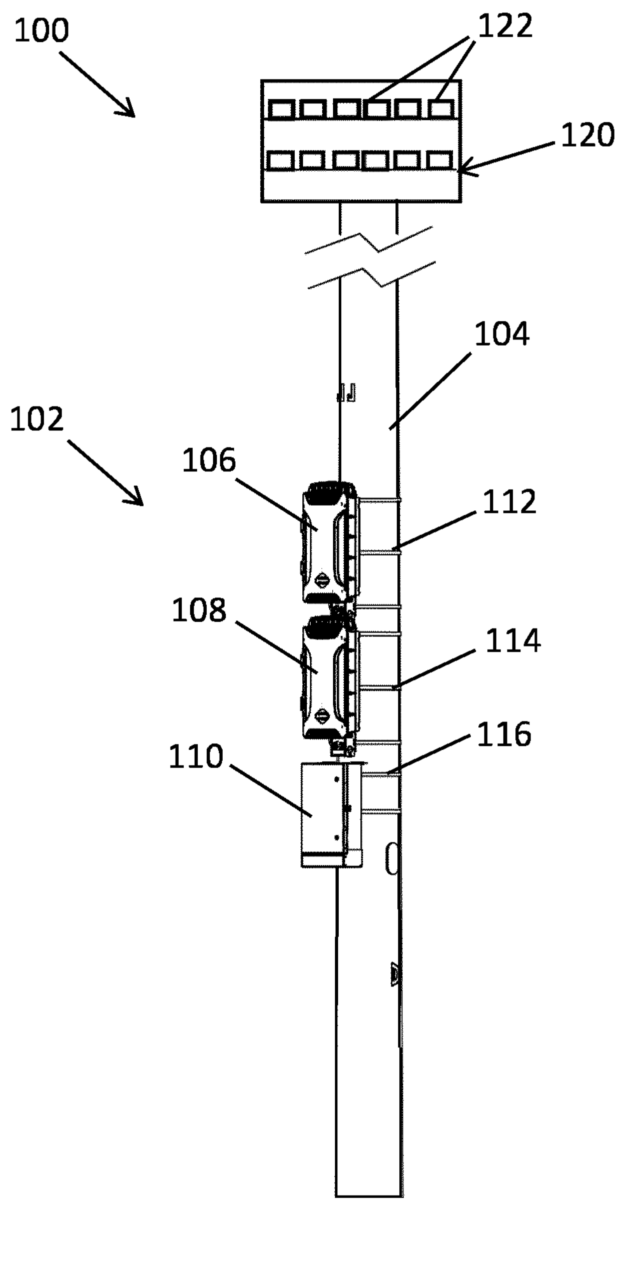


FIG. 1

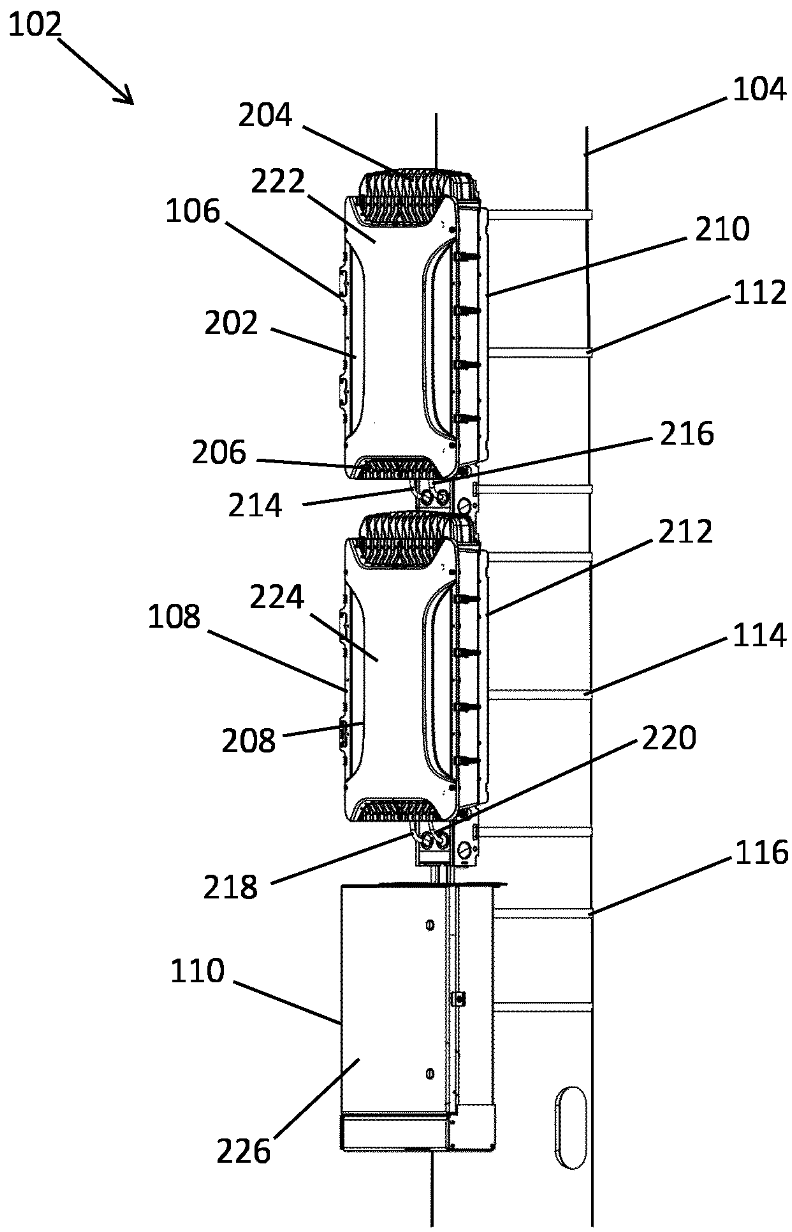

102
204
104
222
106
210
112
202
206
216
214
212
224
108
114
208
220
218
116
110
226

FIG. 2

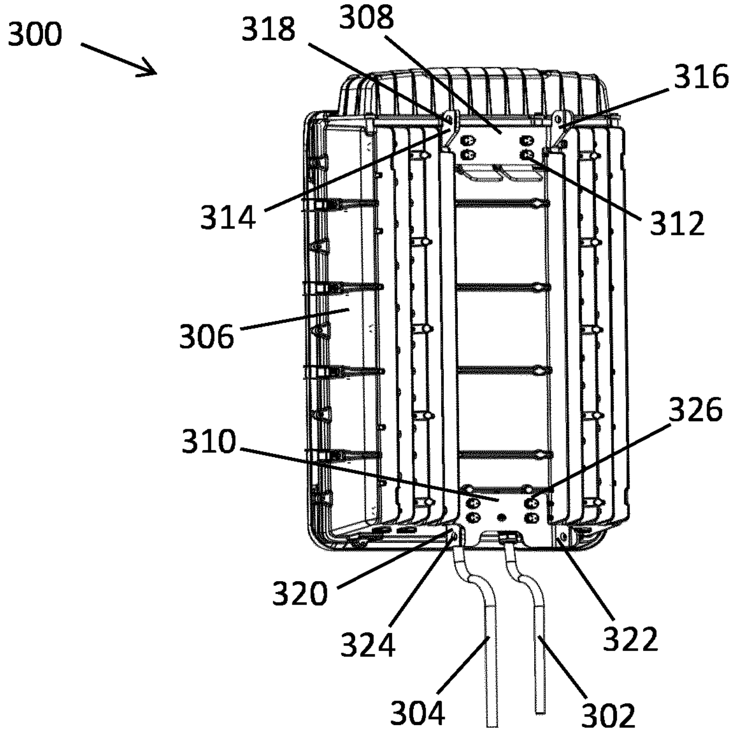

300
318
308
316
314
312
306
310
326
320
324
322
304
302

FIG. 3A

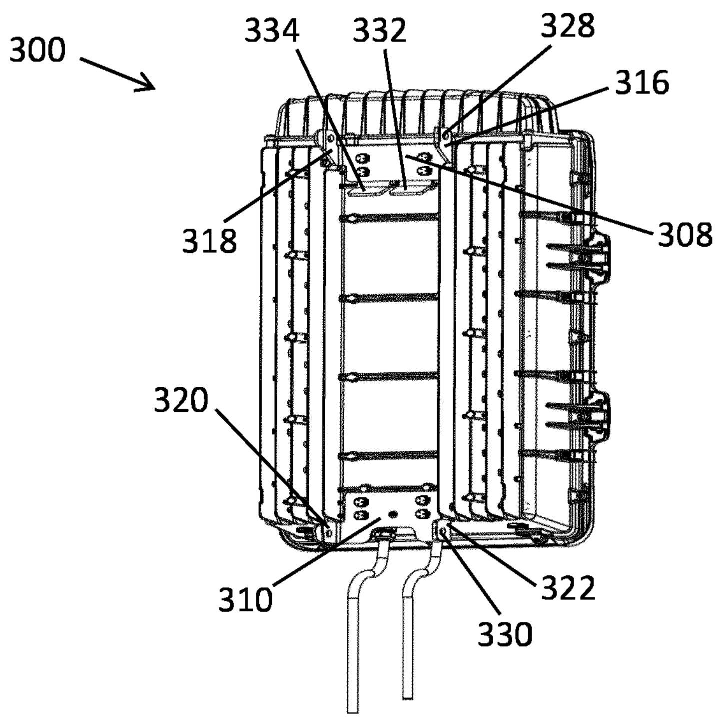

300
334
332
328
316
318
308
320
310
322
330

FIG. 3B

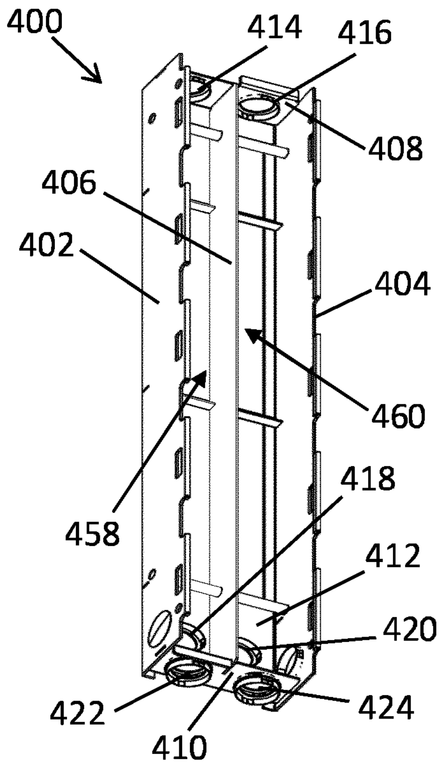


FIG. 4A

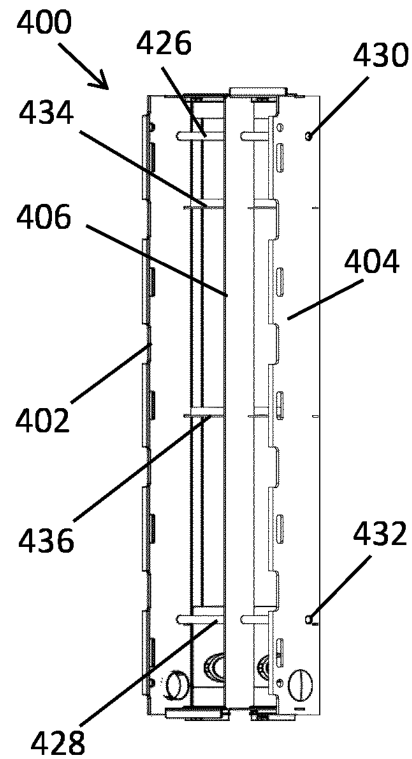


FIG. 4B

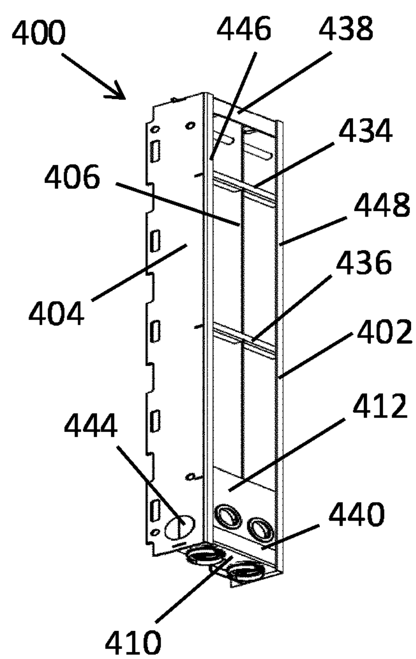


FIG. 4C

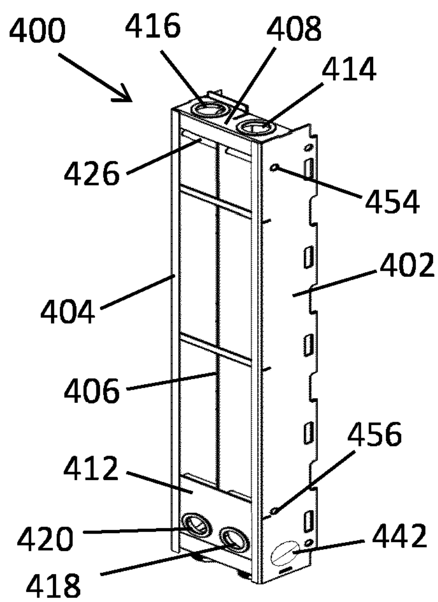


FIG. 4D

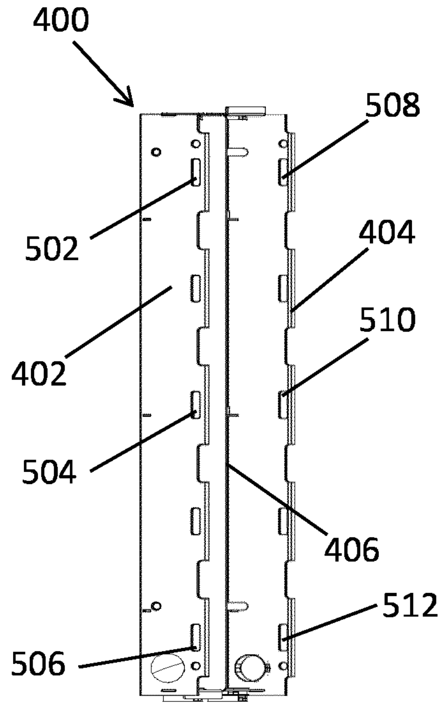


FIG. 5A

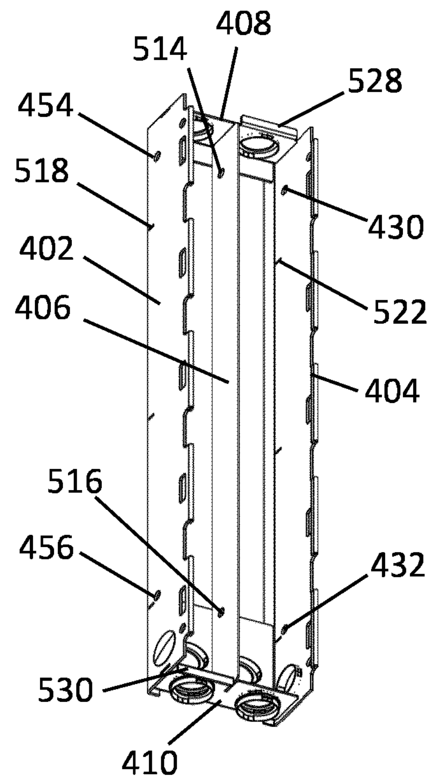


FIG. 5B

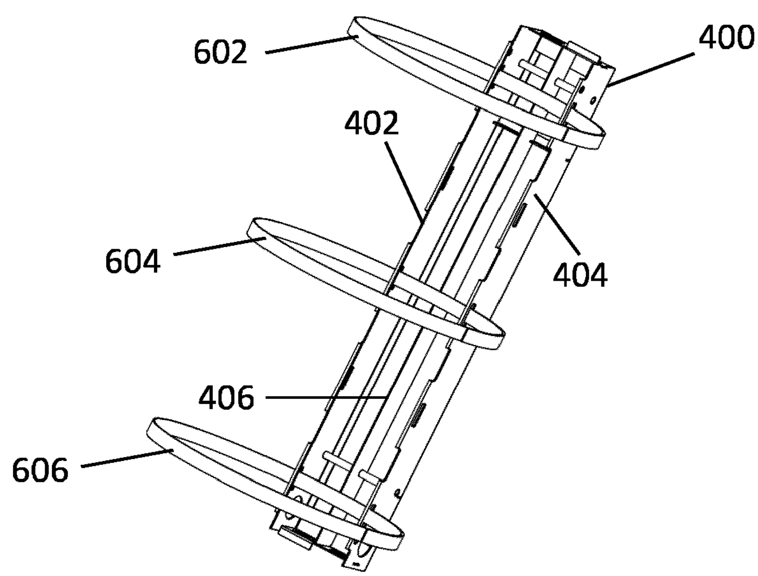


FIG. 6

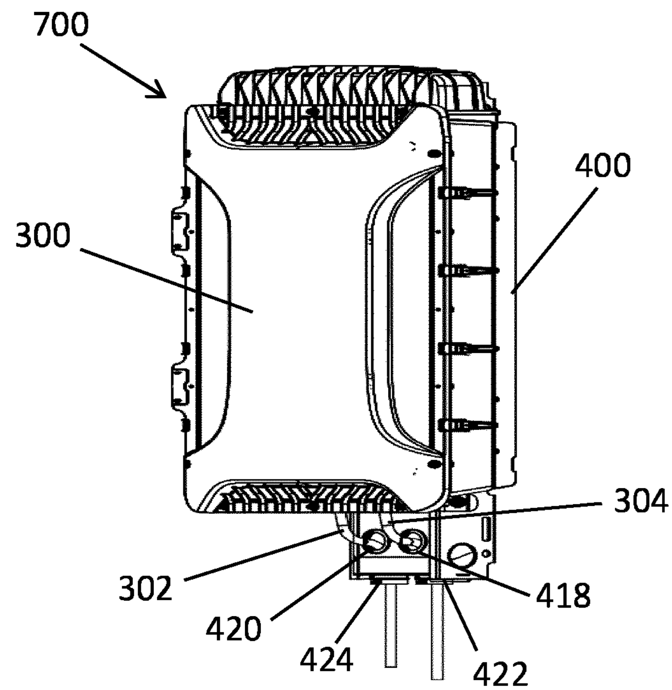
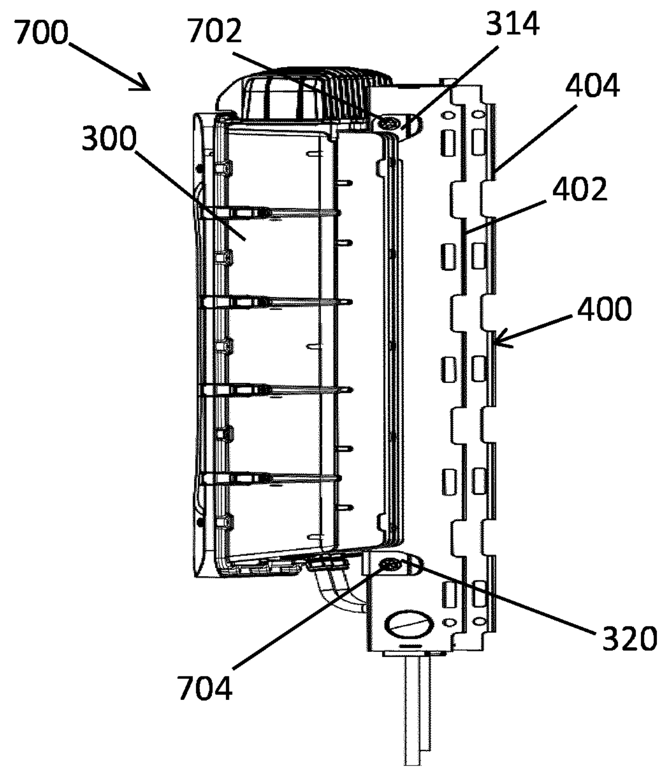

700
400
300
304
302
418
420
424
422

FIG. 7A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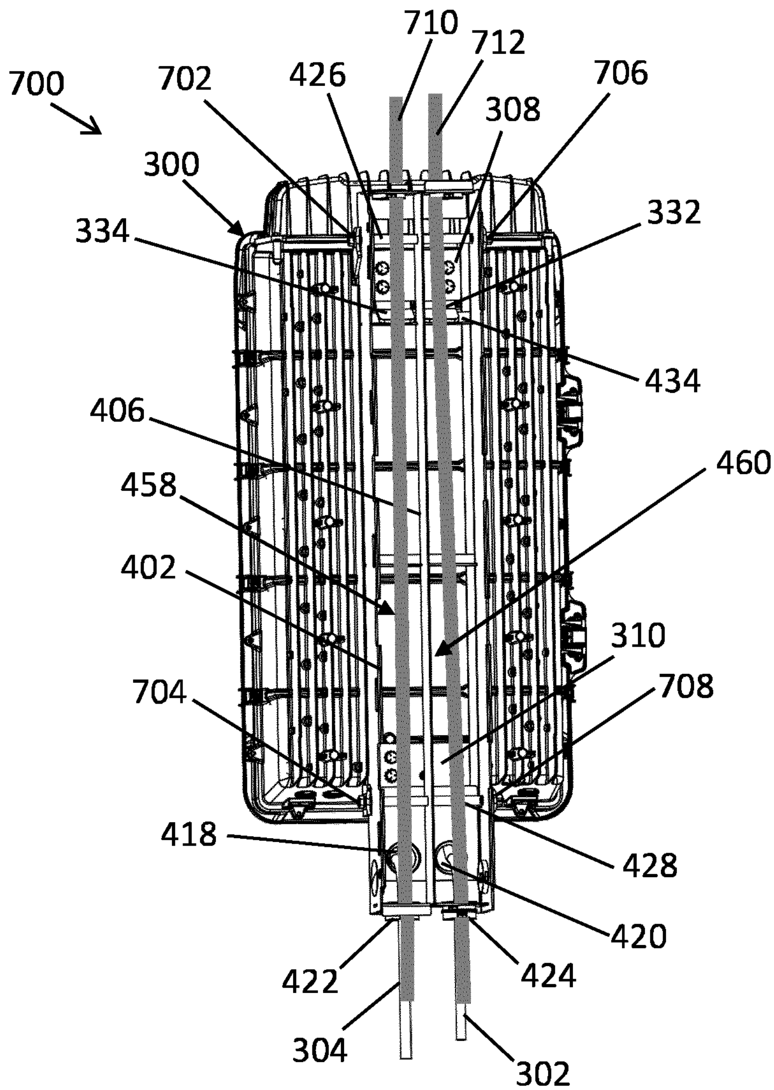
702
314
404
300
402
400
320
704

FIG. 7B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702
426
710
712
308
706
300
334
332
434
406
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458
402
310
704
708
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428
420
422
424
304
302

FIG. 7C

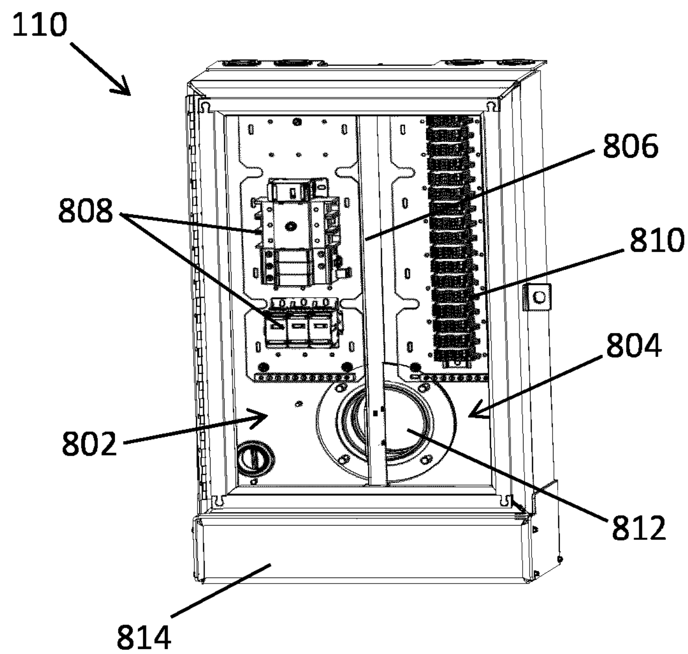
110
806
808
810
804
802
812
814

FIG. 8A

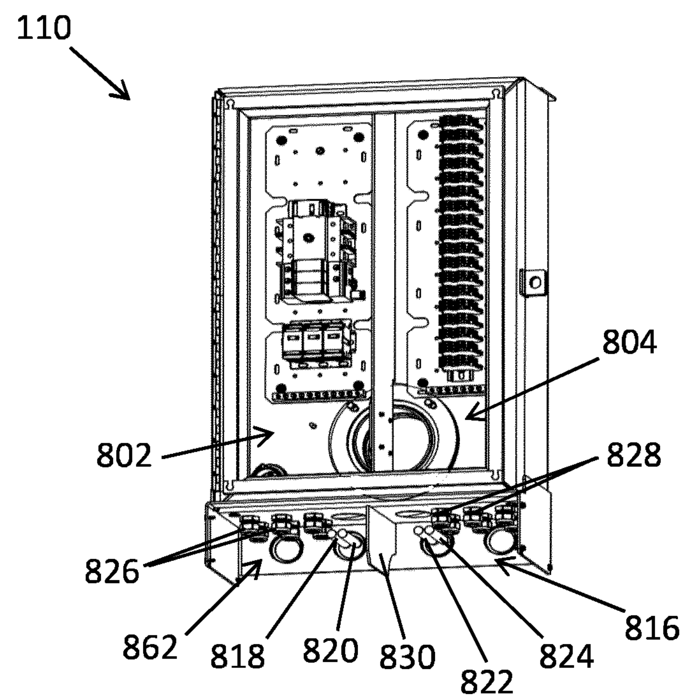
110
804
802
828
826
862
818
820
830
822
824
816

FIG. 8B

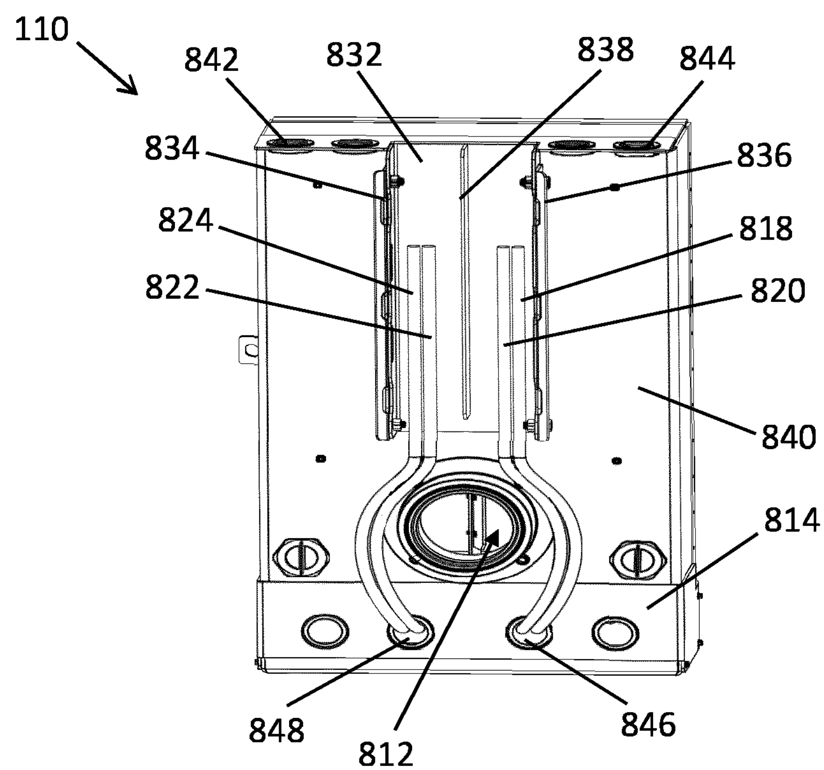
110
842
832
838
844
834
836
824
818
822
820
840
814
848
812
846

FIG. 8C

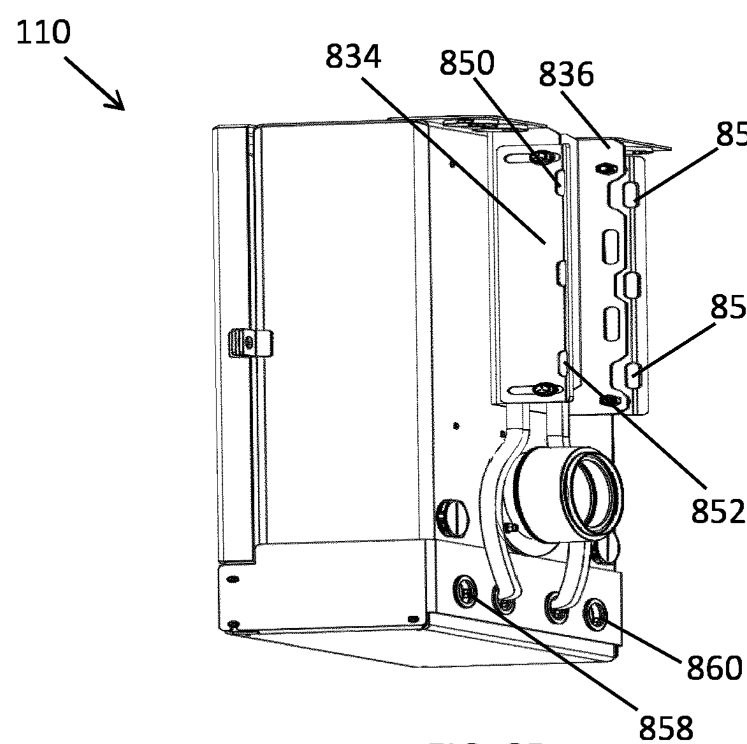
110
834
850
836
854
856
852
860
858

FIG. 8D

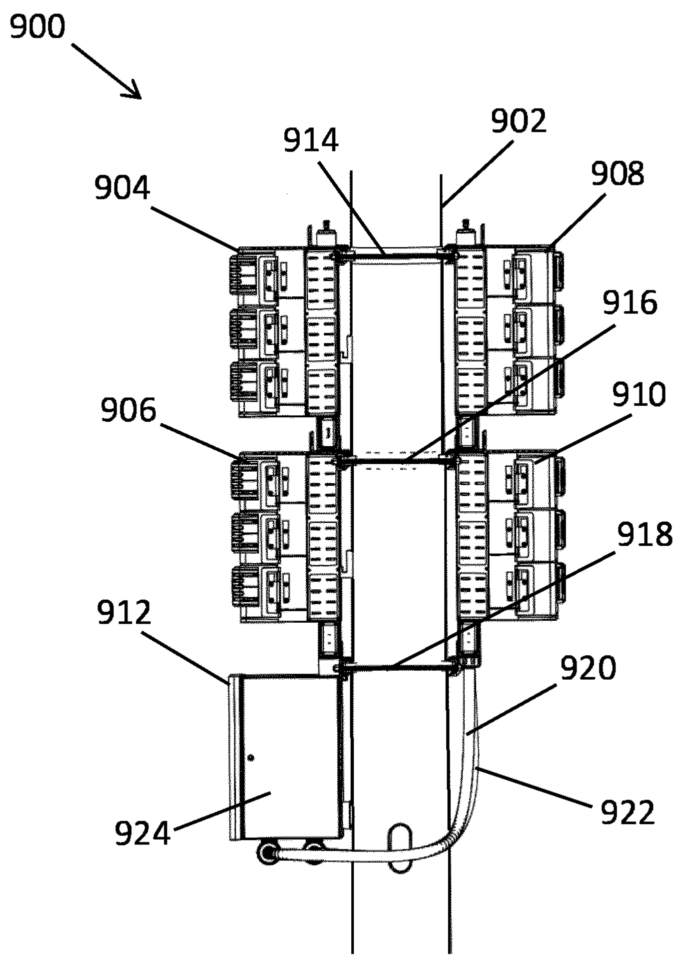

FIG. 9

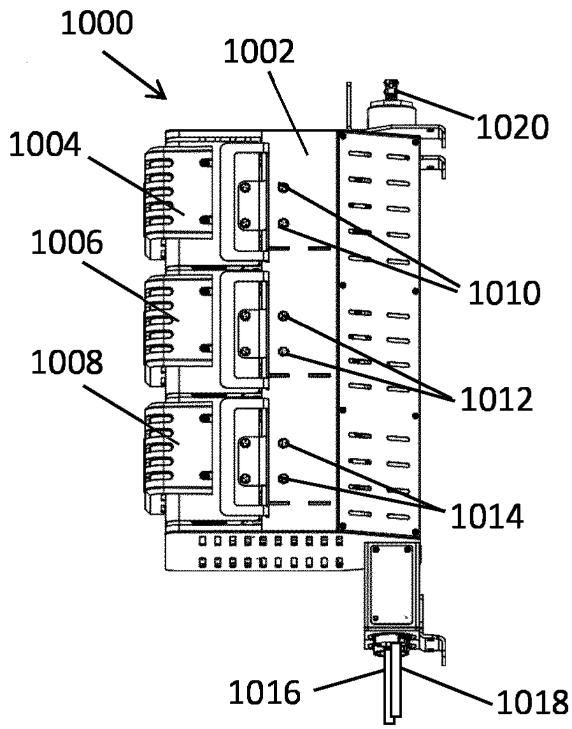

FIG. 10

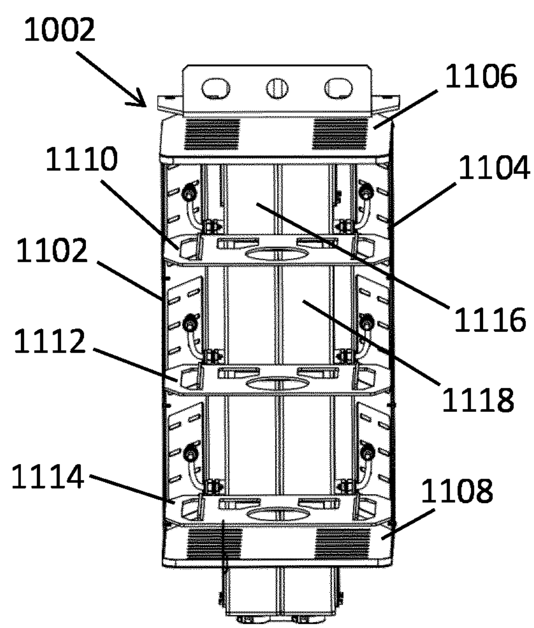

FIG. 11A

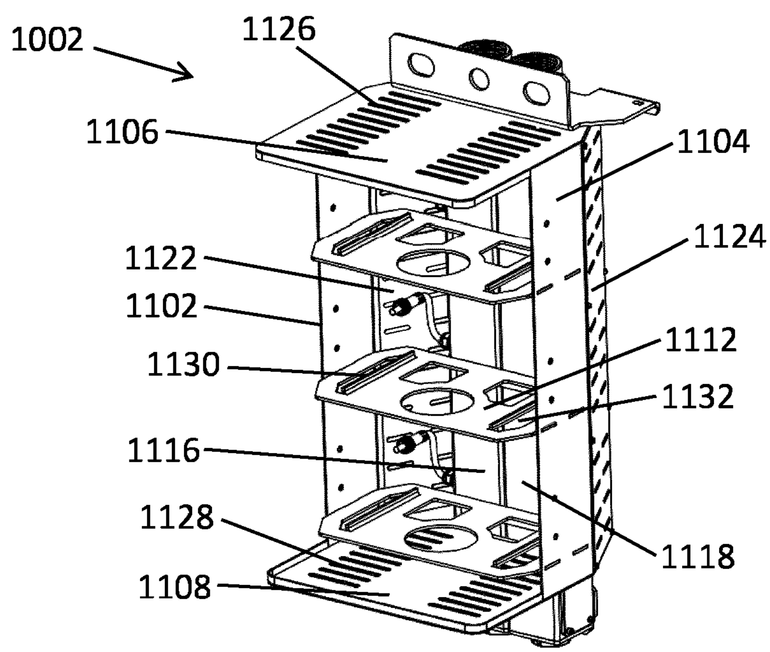

FIG. 11B

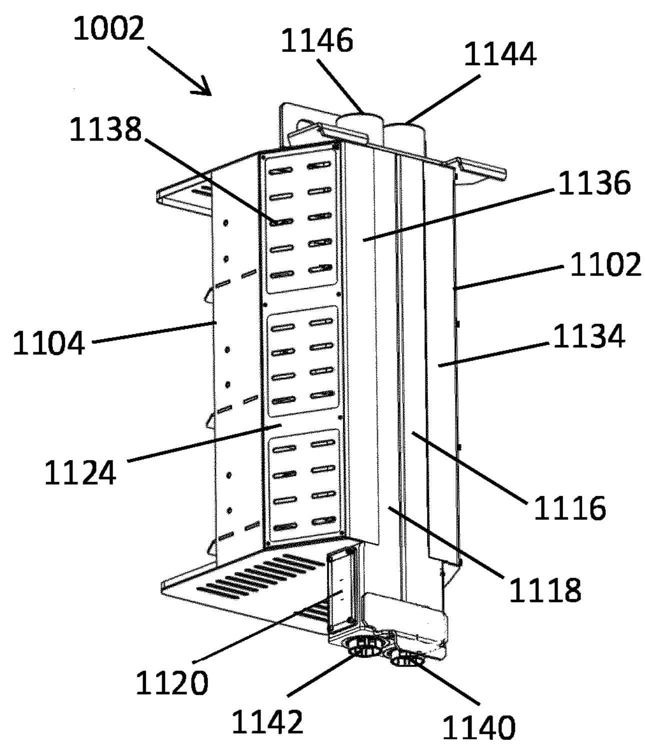

FIG. 11C

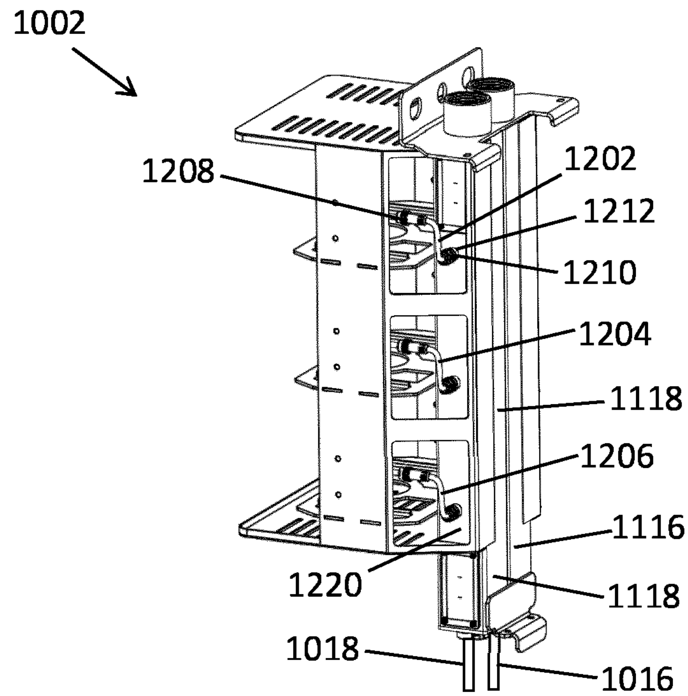

FIG. 12A

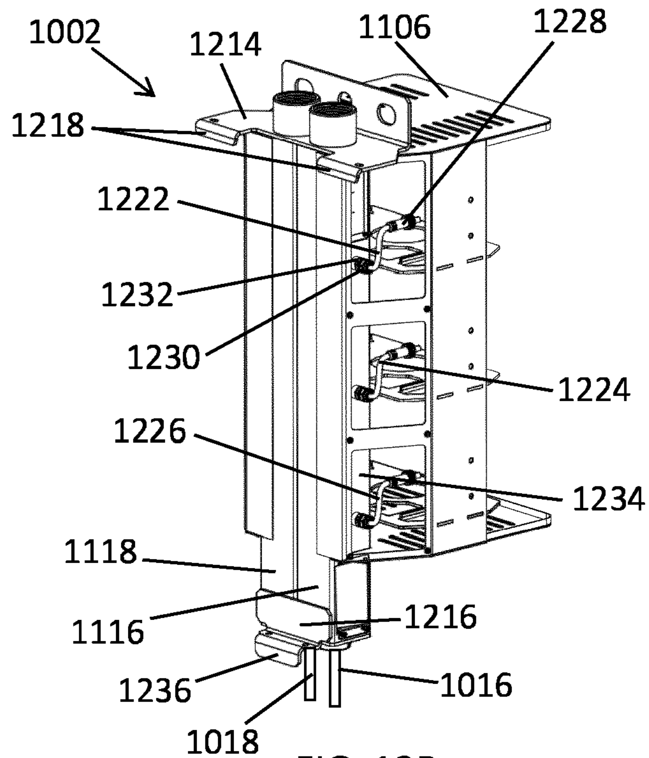

FIG. 12B

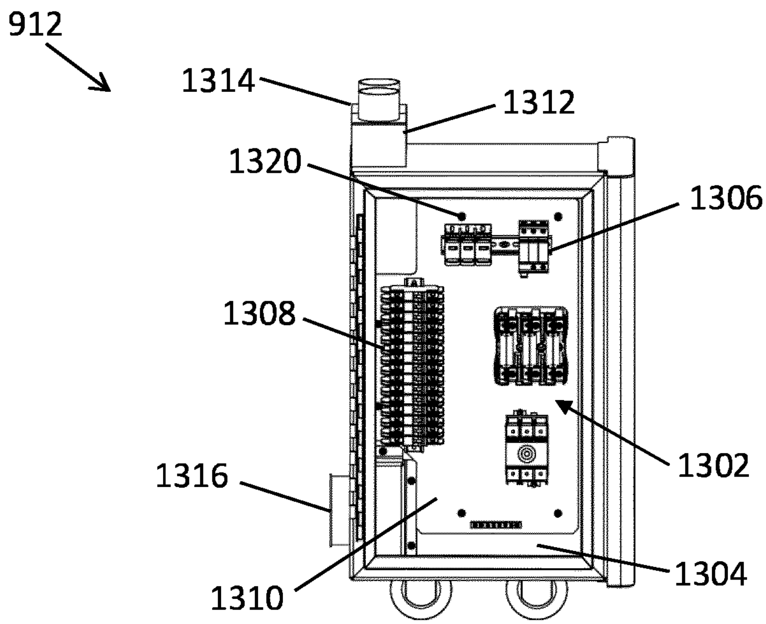

FIG. 13A

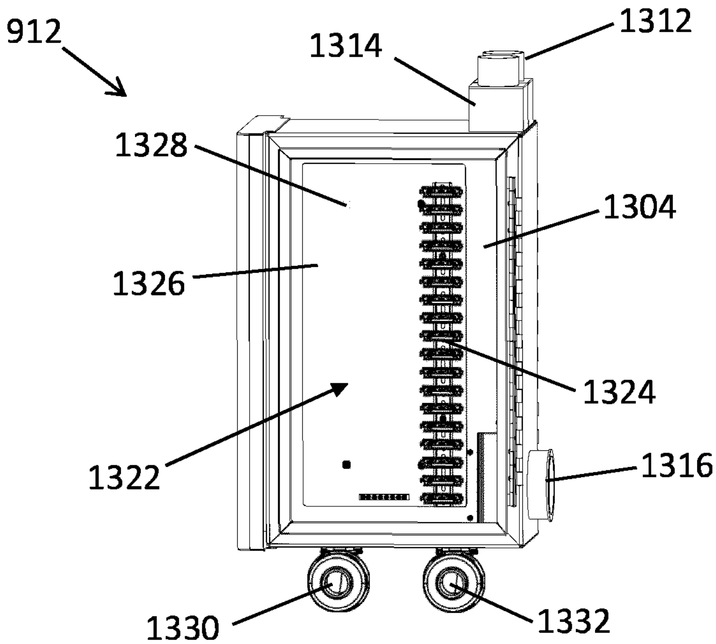

FIG. 13B

REMOTE POWER SYSTEM RACK AND ASSEMBLY

TECHNICAL FIELD

[0001] The present disclosure relates generally to lighting, and more particularly to remote power system assemblies and ladder racks of the remote power system assemblies.

BACKGROUND

[0002] Some outdoor lighting systems (e.g., sports lighting systems) may have support structures, such as poles and frames, that do not support lightheads (e.g., lighting emitting diode (LED) lightheads) that have integrated drivers. In such cases, remotely located power systems may be used to provide power to the lightheads. For example, a remote power system may be attached to a light pole (e.g., close to a base of the light pole) away from the lightheads that are located at a top end of the light pole. In some cases, attaching components of a remote power system to a pole may be challenging, particularly when alternating-current (AC) wires and direct-current (DC) wires should be separated from each other. Thus, a solution that provides for a proper installation of remote power systems and properly installed remote power systems may be desirable.

[0003] U.S. Pat. No. 6,215,069B1 relates to a modular cable management panel assembly includes a panel, a set of rings mounted on a surface of the panel and a cover coupled to the rings. The rings are spaced along the panel longitudinal axis and have openings extending laterally through them. The cover is movable between a closed position extending over and closing the ring openings and an open position spaced from the ring openings to allow cables to pass laterally into and out of the rings. The panel assembly is adaptable for mounting various accessories.

[0004] EP0844713A1 relates to a cable duct made from at least two U-section side modules with grilles to form a Faraday cage, a base and a cover made from a composition material with electromagnetic shielding, cross bars for supporting the cables, and systems for ensuring radio-electrical continuity between the base and cover and between the modules of the duct and the premises in which it is installed. The grille is made from perforated brass with a weight of about 1 kg/sq m, while the radio-electrical continuity between modules is provided by high-frequency joints of copper/nickel. The composition material used for the base and cover is of fiberglass fabric and fire-resistant polyester resin. The assembled duct, apart from the seals is covered with a phenol resin.

[0005] WO2014068382A2 relates to a tamper-resistant cable cover assembly suitable for covering cables that are carried by a cable rack or cable ladder so as to protect such cables against unauthorized tampering or vandalism. The cable cover assembly comprises a cable cover plate which is adapted to engage the cable rack, tray or ladder so as at least partially to sandwich the cables between the cable cover plate and the cable rack, tray or ladder; and connecting means for connecting the cable cover plate to the cable rack, tray or ladder. The cable cover assembly also may include a backing plate which is adapted to engage the cable rack, tray or ladder, the arrangement being such that the cable cover plate engages the cable rack, tray or ladder from one side thereof, while the backing plate engages the cable rack, tray or ladder from an opposite side thereof, so as at least partially to sandwich the cable rack, tray or ladder between the cable cover plate and the backing plate.

SUMMARY

[0006] The present disclosure relates generally to lighting, and more particularly to remote power system assemblies and ladder racks of the remote power system assemblies. In an example embodiment, a mounting rack for attachment of a remote power unit to a structure includes a first side frame, a second side frame, and a separator plate. The first side frame and the separator plate provide an alternating current (AC) wiring pathway therebetween for routing a first electrical cable. The second side frame and the separator plate provide a direct current (DC) wiring pathway therebetween for routing a second electrical cable. The AC wiring pathway and the DC wiring pathway are separated by the separator plate. Further including a bottom plate attached to the first side frame and the second side frame, the bottom plate having a first wire opening aligned with the AC wiring pathway and a second wire opening aligned with the DC wiring pathway, and a top plate attached to the first side frame and the second side frame, the top plate having a third wire opening aligned with the AC wiring pathway and a fourth wire opening aligned with the DC wiring pathway.

[0007] In another example embodiment, a remote power unit assembly includes a remote power unit and a bracket attached to the remote power unit. The remote power unit assembly further includes a mounting rack that includes a first side frame, a second side frame (**404**), and a separator plate. The first side frame and the separator plate provide an AC wiring pathway therebetween for routing a first electrical cable. The second side frame and the separator plate provide a DC wiring pathway therebetween for routing a second electrical cable. The AC wiring pathway and the DC wiring pathway are separated by the separator plate. The bracket is attached to the first side frame and the second side frame. The first side frame and the second side frame each comprise a fastener hole, and wherein the bracket is attached to the first side frame and the second side frame using one or more fasteners extending through the fastener hole of each of the first side frame and the second side frame, and the mounting rack further comprises a hollow tube aligned with the fastener hole of each of the first side frame and the second side frame and wherein the hollow tube extends through a hole in the separator plate.

[0008] These and other aspects, objects, features, and embodiments will be apparent from the following description and the appended claims.

BRIEF DESCRIPTION OF THE FIGURES

[0009] Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0010] FIG. **1** illustrates an outdoor lighting system according to an example embodiment;

[0011] FIG. **2** illustrates a remote power system of the outdoor lighting system of FIG. **1** according to an example embodiment;

[0012] FIGS. **3**A and **3**B illustrate a remote power unit for use in the remote power system of FIG. **2** according to an example embodiment;

[0013] FIGS. 4A-4D illustrate different views of a mounting rack for use in the remote power system of FIG. 2 for attaching the remote power unit of FIG. 3 to a pole according to an example embodiment;

[0014] FIGS. 5A and 5B illustrate the mounting rack of FIGS. 4A-4D with some components omitted for clarity of illustration according to an example embodiment;

[0015] FIG. 6 illustrates the mounting rack of FIGS. 4A-4D with attached mounting bands according to an example embodiment;

[0016] FIGS. 7A-7C illustrate different views of a remote power unit assembly including the remote power unit of FIGS. 3A and 3B and the mounting rack of FIGS. 4A-4D according to an example embodiment;

[0017] FIGS. 8A-8D illustrate different views of a power connection box of the remote power system of FIG. 2 according to an example embodiment;

[0018] FIG. 9 illustrates a remote power system of an outdoor lighting system according to another example embodiment;

[0019] FIG. 10 illustrates a remote power unit assembly for use in the remote power system of FIG. 9 according to an example embodiment;

[0020] FIGS. 11A-11C illustrate different views of a mounting rack of the remote power unit assembly of FIG. 10 used for attaching the remote power unit assembly of FIG. 10 to a pole according to an example embodiment;

[0021] FIGS. 12A and 12B illustrate the mounting rack of FIGS. 11A-4C with side panels removed for clarity of illustration according to an example embodiment; and

[0022] FIGS. 13A and 13B illustrate different views of a power connection box of the remote power system of FIG. 9 according to an example embodiment.

[0023] The drawings illustrate only example embodiments and are therefore not to be considered limiting in scope. The elements and features shown in the drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the example embodiments. Additionally, certain dimensions or placements may be exaggerated to help visually convey such principles. In the drawings, the same reference numerals used in different drawings may designate like or corresponding but not necessarily identical elements.

DETAILED DESCRIPTION OF THE EXAMPLE EMBODIMENTS

[0024] In the following paragraphs, example embodiments will be described in further detail with reference to the figures. In the description, well known components, methods, and/or processing techniques are omitted or briefly described. Furthermore, reference to various feature(s) of the embodiments is not to suggest that all embodiments must include the referenced feature(s).

[0025] FIG. 1 illustrates an outdoor lighting system 100 according to an example embodiment, and FIG. 2 illustrates a remote power system 102 of the outdoor lighting system 100 of FIG. 1 according to an example embodiment. In some example embodiments, the outdoor lighting system 100 includes a lighthead assembly 120 and the remote power system 102. The lighthead assembly 120 may be attached to a pole 104 at a top end portion of the pole 104. The remote power system 102 may be attached to the pole 104 or to another structure (e.g., a wall) at a location distal from the lighthead assembly 120. The remote power system 102 may receive AC power, for example, via an electrical cable (e.g., one or more electrical wires) that is routed up through the pole 104 from below the remote power system 102. The remote power system 102 may generate DC power from the AC power and provide the DC power to the lightheads, such as lightheads 122, of the lighthead assembly 120. For example, the remote power system 102 may provide the DC power to the lightheads of the lighthead assembly 120 via one or more electrical cables connected to the power connection box and routed inside the pole 104. The lightheads of the lighthead assembly 120 may be LED lightheads.

[0026] In some example embodiments, the remote power system 102 may include remote power units 106, 108 and a power connection box 110. The remote power unit 106 is attached to a mounting rack 210, and the remote power unit 108 is attached to a mounting rack 212 (as more clearly shown in FIG. 2). An assembly of the remote power unit 106 and the mounting rack 210 may be attached to the pole 104 by one or more bands (e.g., the band 112) that extend around pole 104. In particular, the remote power unit 106 may be attached to the mounting rack 210 that is attached to the pole 104 by one or more bands. Alternatively or in addition, other attachment structures may be used to attach the mounting rack 210 to the pole 104. An assembly of the remote power unit 108 and the mounting rack 212 may be attached to the pole 104 by one or more bands (e.g., the band 114) that extend around the pole 104. In particular, the remote power unit 108 may be attached to the mounting rack 212 that is attached to the pole 104 by one or more bands. Alternatively or in addition, other attachment structures may be used to attach the mounting rack 210 to the pole 104. The power connection box 110 may be attached to a bracket (more clearly shown in FIG. 8C) that is attached to the pole 104 by one or more bands (e.g., the band 116). Alternatively or in addition, other attachment structures may be used to attach the power connection box 110 to the pole 104.

[0027] In some example embodiments, the remote power unit 106 may include a housing 202 and heat sink structures 204, 206. The heat sink structures 204, 206 may be separate structures from the housing 202 or may be integral portions of the housing 202. The housing 202 may be designed to protect components inside the housing 202 from water exposure. For example, the remote power unit 106 may include one or more drivers (e.g., LED drivers) that receive AC power and generate DC power that is provided to some lightheads of the lighthead assembly 120. The driver(s) of the remote power unit 106 may be accessed by opening a door 222 of the housing 202.

[0028] In some example embodiments, the remote power unit 108 may include a housing 208 and heat sink structures similar to those of the remote power unit 106. The housing 208 may be designed to protect components inside the housing 208 from water exposure. For example, the remote power unit 108 may include one or more drivers (e.g., LED drivers) that receive AC power and generate DC power that is provided to some lightheads of the lighthead assembly 120. The driver(s) of the remote power unit 108 may be accessed by opening a door 224 of the housing 208.

[0029] In some example embodiments, electrical cables 214, 216 may extend from the remote power unit 106 to the power connection box 110. For example, the electrical cable 214 may be used for AC power connection between the power connection box 110 and the remote power unit 106, and electrical cable 216 may be used for DC power connection between the remote power unit **106** and the power connection box **110**. To illustrate, AC power (e.g., from power utility) may be provided to the remote power unit **106** over the electrical cable **214** via the power connection box **110**. The remote power unit **106** may generate DC power from the AC power and may provide the DC power to some lightheads of the lighthead assembly **120** over the electrical cable **216** via the power connection box **110** and one or more electrical cables connected to the power connection box **110** and routed to the lighthead assembly **120** through the inside of the pole **104**. The electrical cables **214**, **216** may be routed to the power connection box **110** through wiring pathways provided by the mounting racks **210**, **212**. The electrical cables **214**, **216** may be routed separated from each other by separator plates of the mounting racks **210**, **212** in a manner described below with respect to a mounting rack **400** of FIGS. **4A-4D**.

[0030] In some example embodiments, electrical cables **218**, **220** may extend from the remote power unit **108** to the power connection box **110**. For example, the electrical cable **218** may be used for AC power connection between the power connection box **110** and the remote power unit **108**, and electrical cable **220** may be used for DC power connection between the remote power unit **108** and the power connection box **110**. To illustrate, AC power (e.g., from power utility) may be provided to the remote power unit **108** over the electrical cable **218** via the power connection box **110**. The remote power unit **108** may generate DC power from the AC power and may provide the DC power to some lightheads of the lighthead assembly **120** over the electrical cable **220** via the power connection box **110** and one or more electrical cables connected to the power connection box **110** and routed to the lighthead assembly **120** through the inside of the pole **104**. The electrical cables **218**, **220** may be routed to the power connection box **110** through wiring pathways provided by the mounting rack **212**. The electrical cables **218**, **220** may be routed separated from each other by a separator plate of the mounting rack **212** in a manner described below with respect to a mounting rack **400** of FIGS. **4A-4D**.

[0031] In some example embodiments, the electrical cables **214**, **218** may be routed behind and/or inside the power connection box **110** separated from the electrical cables **216**, **220** to provide separation of AC and DC power wirings. The power connection box **110** may include separate compartments for AC and DC power connections as explained in more detail below with respect to FIGS. **8A-8D**. The compartments of the power connection box **110** may be accessed by opening a door **226** of the power connection box **110**.

[0032] Because the remote power system **102** provides DC power to the lighthead assembly **120** while remotely located from the lighthead assembly **120**, the lightheads of the lighthead assembly **120** do not need to include local power supplies. By eliminating the need for the lightheads to incorporate a driver, the remote power system **102** enables replacement of non-LED lightheads with LED lightheads. By providing separate wiring pathways and compartments for AC and DC power connections, the remote power system **102** improves the separation of AC and DC power, which can, for example, reduce signal interference. The separation of the remote power units **106**, **108** from each other by at least the portion of the mounting rack **210** that is below the remote power unit **106** may provide adequate spacing for efficient dissipation of heat from the remote power units **106**, **108**.

[0033] Although two remote power units are shown in FIGS. **1** and **2**, in some alternative embodiments, the remote power system **102** may include more or fewer remote power units without departing from the scope of this disclosure. In some alternative embodiments, one or more of the remote power units **106**, **108** and the power connection box **110** may be attached to the pole **104** at different locations than shown without departing from the scope of this disclosure. In some example embodiments, electrical cables described above with respect to AC power may be used for DC power connection while the electrical cables described above with respect to DC power may be used for AC power connection without departing from the scope of this disclosure.

[0034] FIGS. **3**A and **3**B illustrate a remote power unit **300** for use in the remote power system **102** of FIG. **2** according to an example embodiment. For example, the remote power unit **300** may correspond to each one of the remote power units **106** and **108**. In some example embodiments, the remote power unit **300** may include a housing **306** that holds, for example, one or more drivers (not shown), such as LED drivers. Electrical cables **302**, **304** may be electrically connected to (i.e., in electrical connection with) the drivers inside the housing **306** or via connectors attached to the housing **306**. To illustrate, the electrical cable **302** may be used for AC power connection, and the electrical cable **304** may be used for DC power connection. For example, the electrical cables **302** and **304** may correspond, respectively, to the electrical cables **214** and **216** and, respectively to the electrical cables **218**, **220** shown more clearly in FIG. **2**.

[0035] In some example embodiments, brackets **308**, **310** may be attached to the housing **306** of the remote power unit **300**. The bracket **308** may be used to attach the remote power unit **300** to a mounting rack, such as the mounting racks **210**, **212** shown in FIG. **2**. The brackets **308** may be attached to the housing **306** on a back side of the remote power unit **300** using one or more fasteners (e.g., screws), such as the fastener **312**. The bracket **308** may include attachment tabs **314**, **316** that extend away from the housing **306**. The tab **314** may include an attachment hole **318** for extending a fastener therethrough, and the tab **316** may include an attachment hole **328** for extending the same or another fastener therethrough. The bracket **308** may also include retention tabs **332**, **334** that extend angularly away from the housing **306** and designed to be positioned on a horizontal structure (e.g., a support rung) of a mounting rack to help retain the remote power unit **300** attached to the mounting rack.

[0036] In some example embodiments, the bracket **310** may be attached to the housing **306** on the back side of the remote power unit **300** using one or more fasteners (e.g., screws), such as the fastener **326**. The bracket **310** may include tabs **320**, **322** that extend away from the housing **306**. The tab **320** may include an attachment hole **324** for extending a fastener therethrough for attaching the remote power unit **300** to a mounting rack (e.g., the mounting rack **210** of FIG. **2**), and the tab **322** may include an attachment hole **330** for extending the same or another fastener therethrough.

[0037] In some alternative embodiments, the remote power unit **300** may have a different shape than shown without departing from the scope of this disclosure. In some alternative embodiments, the brackets **308**, **310** may have different shapes and/or may be attached at different locations than shown without departing from the scope of this disclosure. In some alternative embodiments, the bracket **310** may include retention tabs similar to the retention tabs **332**, **334** of the bracket **308** without departing from the scope of this disclosure. In some alternative embodiments, one or more of the retention tabs **332**, **334** of the bracket **308** may be omitted without departing from the scope of this disclosure. In some alternative embodiments, the brackets **308**, **310** may be attached to the housing **306** of the remote power unit **300** using other means than shown without departing from the scope of this disclosure. In some alternative embodiments, one or more brackets in addition to the brackets **308**, **310** may be attached to the housing **306** without departing from the scope of this disclosure. In some alternative embodiments, one of the brackets **308**, **310** may be omitted without departing from the scope of this disclosure.

[0038] FIGS. **4**A-**4**D illustrate different views of a mounting rack **400** for use in the remote power system **102** of FIG. **2** for attaching the remote power unit **300** of FIG. **3** to the pole **104** of FIG. **1** according to an example embodiment, and FIGS. **5**A and **5**B illustrate the mounting rack **400** of FIGS. **4**A-**4**D with some components omitted for clarity of illustration according to an example embodiment. In some example embodiments, the mounting rack **400** includes side frame **402**, **404**, and a separator plate **406** that extends longitudinally along with and between the side frames **402**, **404**. The side frame **402** and the separator plate **406** may provide a wiring pathway **458** for routing one or more electrical cables, and the side frame **404** and the separator plate **406** may provide another wiring pathway **460** for routing another one or more electrical cables. For example, the wiring pathway **458** may be used for routing one or more electrical cables that are used for AC power connection, and the wiring pathway **460** may be used for routing one or more electrical cables that are used for DC power connection. The separator plate **406** is positioned to separate one or more electrical cables that are routed in the wiring pathway **458** from one or more electrical cables that are routed in the wiring pathway **460**.

[0039] In some example embodiments, the mounting rack **400** may include a top plate **408**, a bottom plate **410**, and a front plate **412** proximal to the bottom plate **410**. The top plate **408** may include wire openings **414**, **416**. For example, the wire opening **414** may be aligned with the wiring pathway **458**, and the wire opening **416** may be aligned with the wiring pathway **460**. The front plate **412** may include wire openings **418**, **420**. For example, the wire opening **418** may be aligned with the wiring pathway **458**, and the wire opening **420** may be aligned with the wiring pathway **460**. The bottom plate **410** may include wire openings **422**, **424**. For example, the wire opening **422** may be aligned with the wiring pathway **458**, and the wire opening **424** may be aligned with the wiring pathway **460**.

[0040] In some example embodiments, the mounting rack **400** may include support rungs **434**, **436** that extend between the side frame **402** and the side frame **404**. The support rungs **434**, **436** may provide structural support to the mounting rack **400**. For example, the support rung **434** may be attached to the side frame **402**, where an end portion (e.g., an end tab) of the support rung **434** is inserted in a slot **518** shown in FIG. **5**B. The support rung **434** may be also attached to the side frame **404**, where another end portion (e.g., an end tab) of the support rung **434** is inserted in a slot **522** shown in FIG. **5**B. In some alternative embodiments, the support rung **434** may be attached to the side frames **402**, **404** in a different manner (e.g., fasteners and/or welding) without departing from the scope of this disclosure. The support rung **436** may be attached to the side frames **402**, **404** in the manner described with respect to the support rung **434**.

[0041] In some example embodiments, the mounting rack **400** includes hollow tubes **426**, **428** that are designed to receive fasteners that are used to attach the remote power unit **300** to the mounting rack **400**. The hollow tubes **426**, **428** may be fully or partially hollow. The hollow tubes **426**, **424** may also be threaded to receive a threaded fastener. The hollow tube **426** may be aligned with a fastener hole **454** in the side frame **402** (as more clearly shown in FIG. **4**D) such that a fastener can extend through the fastener hole **454** and the hollow tube **426**. The hollow tube **426** may also be aligned with a fastener hole **430** in the side frame **404** (as more clearly shown in FIG. **4**B) such that the same fastener or another fastener can extend through the fastener hole **430** and the hollow tube **426**. The hollow tube **430** may also extend through a hole **514** (shown in FIG. **5**B) of the separator plate **406**, which may restrict the movement of the separator plate **406** while retaining the hollow tube **426** in position.

[0042] In some example embodiments, the hollow tube **428** may be aligned with a fastener hole **456** in the side frame **402** (as more clearly shown in FIG. **4**D) such that a fastener can extend through the fastener hole **456** and the hollow tube **428**. The hollow tube **428** may also be aligned with a fastener hole **432** in the side frame **404** (as more clearly shown in FIG. **4**B) such that the same fastener or another fastener can extend through the fastener hole **432** and the hollow tube **428**. The hollow tube **432** may also extend through a hole **516** (shown in FIG. **5**B) of the separator plate **406**, which may restrict the movement of the separator plate **406** while retaining the hollow tube **428** in position.

[0043] In some example embodiments, the top plate **408** may include a downward flange **438**, and the bottom plate **410** may include an upward flange **440**. The side frame **402** may include a flange **446**, and the side frame **404** may include a flange **448**. For example, the flanges **438**, **440**, may help retain the separator plate **406** in place, and the flanges **446**, **448** may help retain the support rungs **434**, **436** in place.

[0044] In some example embodiments, the side frame **402** may include a knockout piece **442**, and the side frame **404** may include a knockout piece **444**. For example, the knockout piece **442** may be removed to provide an additional wire opening to the wiring pathway **458**, and the knockout piece **444** may be removed to provide an additional wire opening to the wiring pathway **460**.

[0045] In some example embodiments, one or more electrical cables may extend through the wire opening **414**, the wiring pathway **458**, and the wire opening **422**, and one or more electrical cable may extend through the wire opening **416**, the wiring pathway **460**, and the wire opening **424**. For example, considering the mounting rack **400** as corresponding to the mounting rack **212** of FIG. **2**, the electrical cable **214** may be routed be through the wire opening **414**, the wiring pathway **458**, and the wire opening **422**, and the electrical cable **216** may be routed be through the wire opening **416**, the wiring pathway **460**, and the wire opening **424** separated from the electrical cable **214** by the separator plate **406**.

[0046] In some example embodiments, the one or more electrical cables may extend through the wire opening **418**, the wiring pathway **458**, and the wire opening **422**, and one or more electrical cable may extend through the wire opening **420**, the wiring pathway **460**, and the wire opening **424**. For example, considering the mounting rack **400** as corresponding to the mounting rack **210** of FIG. **2**, the electrical cable **214** (shown in FIG. **2**) may be routed be through the wire opening **418**, the wiring pathway **458**, and the wire opening **422**, and the electrical cable **216** (shown in FIG. **2**) may be routed be through the wire opening **420**, the wiring pathway **460**, and the wire opening **424** separated from the electrical cable **214** by the separator plate **406**. As another example, considering the mounting rack **400** as corresponding to the mounting rack **212** of FIG. **2**, the electrical cable **218** (shown in FIG. **2**) may be routed be through the wire opening **418**, the wiring pathway **458**, and the wire opening **422**, and the electrical cable **220** (shown in FIG. **2**) may be routed be through the wire opening **420**, the wiring pathway **460**, and the wire opening **424** separated from the electrical cable **218** by the separator plate **406**.

[0047] Referring to FIGS. **5**A and **5**B, in some example embodiments, the top plate **408** may include a tab **528**, and the bottom plate **410** may include a tab **530**. The tabs **528**, **530** may be used in the alignment of the mounting rack **400** with another one of the mounting rack **400**. For example, considering the mounting racks **210** and **212** of FIG. **2** as corresponding to the mounting rack **400**, the tab **528** of the top plate **408** of the mounting rack **212** and the tab **530** of the bottom plate **410** of the mounting rack **210** may help align the mounting rack **210** with the mounting rack **212**.

[0048] In some example embodiments, the side frame **402** may include slots **502**, **504**, **506**, and the side frame **404** may include slots **508**, **510**, **512**. For example, the slots **502** and **508** may be aligned with each other, the slots **504** and **510** may be aligned with each other, and the slots **512** and **514** may be aligned with each other. The slots **502-512** may be used for attaching the mounting rack **400** to the pole **104** of FIGS. **1** and **2** using mounting bands.

[0049] To illustrate, FIG. **6** illustrates the mounting rack **400** with attached mounting bands **602**, **604**, **606** according to an example embodiment. Referring to FIGS. **5**A-**6**, in some example embodiments, the mounting band **602** may be inserted through the slot **502** of the side frame **402** and through the slot **508** of the side frame **404**. The mounting band **604** may be inserted through the slot **504** of the side frame **402** and through the slot **510** of the side frame **404**. The mounting band **606** may be inserted through the slot **506** of the side frame **402** and through the slot **512** of the side frame **404**. The mounting bands **602-606** may be used to attach the mounting rack **400** to the pole **104** shown in FIGS. **1** and **2**.

[0050] In some example embodiments, the mounting rack **400** may be made from sheet metal, such as aluminum and/or steel sheet metal, using methods such as cutting, bending, welding, etc. The mounting bands **602-606** may also be made from aluminum and/or steel sheet metal using methods such as cutting, bending, welding, etc.

[0051] Referring to FIGS. **4**A-**6**, in some alternative embodiments, the hollow tubes **426**, **436** and the support rungs **434**, **436** may be at different locations than shown without departing from the scope of this disclosure. In some alternative embodiments, the mounting rack **400** may include more or fewer hollow tubes and support rungs than shown without departing from the scope of this disclosure. In some alternative embodiments, the separator plate **406** may be at a different location relative to the side frames **402**, **404** without departing from the scope of this disclosure. In some alternative embodiments, the side frames **402**, **404** of the mounting rack **400** may include more or fewer of the slots **502-512** than shown without departing from the scope of this disclosure. In some alternative embodiments, one or more elements (e.g., the side frame **402**, the side frame **404**, etc.) may have a different shape than shown without departing from the scope of this disclosure. In some alternative embodiments, the mounting rack **400** may include more or fewer wire holes than shown without departing from the scope of this disclosure. In some alternative embodiments, some of the elements of the mounting rack **400** may be attached in a different manner than shown without departing from the scope of this disclosure.

[0052] FIGS. **7**A-**7**C illustrate different views of a remote power unit assembly **700** including the remote power unit **300** of FIGS. **3**A and **3**B and the mounting rack **400** of FIGS. **4**A-**4**D according to an example embodiment. Referring to FIGS. **3**A-**7**C, in some example embodiments, the electrical cable **304** may be routed through the wire opening **418** of the front plate **412**, the wiring pathway **458**, and the wire opening **422** of the bottom plate **410** of the mounting rack **400**. The electrical cable **302** may be routed through the wire opening **420** of the front plate **412**, the wiring pathway **460**, and the wire opening **424** of the bottom plate **410** of the mounting rack **400**.

[0053] In some example embodiments, electrical cable **710** may extend through the wire opening **414**, the wiring pathway **458**, and the wire opening **422**, and electrical cable **712** may extend through the wire opening **416**, the wiring pathway **460**, and the wire opening **424**. The electrical cables **710** and **712** (shown more clearly in FIG. **7**C) are routed through the wiring pathways **458** and **460**, respectively, separated from each other by the separator plate **406**. Considering the mounting rack **212** of FIG. **2** as corresponding to the mounting rack **400**, the electrical cable **710** may correspond to the electrical cable **216**, and the electrical cable **712** may correspond to the electrical cable **214**.

[0054] In some example embodiments, a fastener **702** may extend through the hollow tube **426** attaching the bracket **308** to the mounting rack **400**, thereby attaching the remote power unit **300** to the mounting rack **400** as more clearly shown in FIGS. **7**B and **7**C. For example, the fastener **702** (e.g., a screw) may extend through the fastener hole **430** in the side frame **402**, the fastener hole **454** in the side frame **404**, and the hollow tube **426**, where the fastener **702** is secured by a nut **706**. The retention tabs **332**, **334** of the bracket **308** may be positioned on the support rung **434** of the mounting rack **400** to help retain the remote power unit **300** attached to the mounting rack **400**.

[0055] In some example embodiments, a fastener **704** may extend through the hollow tube **428** attaching the bracket **310** to the mounting rack **400**, thereby attaching the remote power unit **300** to the mounting rack **400** as more clearly shown in FIGS. **7**B and **7**C. The fastener **704** (e.g., a screw) may extend through the fastener hole **432** in the side frame

402, the fastener hole **456** in the side frame **404**, and the hollow tube **428**, where the fastener **704** is secured by a nut **708**.

[0056] In some alternative embodiments, the electrical cables **710**, **712** may be omitted without departing from the scope of this disclosure. In some alternative embodiments, the brackets **308**, **310** may be attached to the mounting rack **400** using means other than the fasteners **702**, **704** without departing from the scope of this disclosure.

[0057] FIGS. **8**A-**8**D illustrate different views of the power connection box **110** of the remote power system **102** of FIG. **2** according to an example embodiment. In some example embodiments, the power connection box **110** may have an AC compartment **802** and a DC compartment **804**. The AC compartment **802** and the DC compartment **804** may be separated from each other by a separator wall **806**. The AC compartment **802** may contain components **808** such as surge protector, circuit breakers, etc. The DC compartment **802** may include components **810** such as wire connectors, etc. The power connection box **110** may be made from sheet metal, such as aluminum and/or steel sheet metal, using methods such as cutting, bending, welding, etc.

[0058] In some example embodiments, the power connection box **110** may include a wire opening **812** for routing electrical cables. For example, the power connection box **110** may be mounted to the pole **104** of FIGS. **1** and **2** such that the wire opening **812** is aligned with an opening in the pole **104** for routing electrical cables between the power connection box **110** and the inside of the pole **104**. The separator wall **806** may enable the routing of AC power cables and DC power cables spaced from each other through the wire opening **812**.

[0059] In some example embodiments, the power connection box **110** includes a wiring compartment **814** that has an AC wiring section **862** and a DC wiring section **816** that are separated from each other by a separator structure **830** (e.g., a plate). The wiring compartment **814** is shown in FIG. **8**B with the cover removed for clarity of illustration. For example, electrical cables **818**, **820**, which may be used for AC power connection between the power connection box **110** and the remote power unit **300** shown in FIGS. **7**A-**7**C, may be routed into the AC wiring section **862** through a wire opening **846**. For example, in FIGS. **8**A-**8**D, the electrical cables **818**, **820** may each be a segment of or otherwise connected to a respective one of the electrical cables **304**, **710** (shown in FIG. 7C). The electrical cables **818**, **820** may be routed between the AC wiring section **862** and the AC compartment **802** of the power connection box **110** through a respective one of the wire grips **826**.

[0060] In some example embodiments, electrical cables **822**, **824**, which may be used for DC power connection between the power connection box **110** and the remote power unit **300** shown in FIGS. **7**A-**7**C, may be routed into the DC wiring section **816** through a wire opening **848**. For example, in FIGS. **8**A-**8**D, the electrical cables **822**, **824** may each be a segment of or otherwise connected to a respective one of the electrical cables **302**, **712** (shown in FIG. **7**C). The electrical cables **822**, **824** may be routed between the DC wiring section **816** and the DC compartment **804** of the power connection box **110** through a respective one of the wire grips **828**.

[0061] In some example embodiments, the electrical cables **818**, **820** may be routed separated from the electrical cables **822**, **824** on a back side of the power connection box **110** as more clearly shown in FIG. **8**C. To illustrate, a mounting bracket **832** that includes sidewalls **834**, **836** and separator wall **838** may be attached to a backwall **840** of the power connection box **110**, where the electrical cables **818**, **820** may be routed between the sidewall **836** and the separator wall **838** and where the electrical cables **822**, **824** may be routed between the sidewall **834** and the separator wall **838**.

[0062] In some example embodiments, power connection box **110** may include wire openings **842** and **844** that may be used to route additional electrical cables between the power connection box **110** and other remote power units. For example, an electrical cable that may be used in AC power connection may be routed through the wire opening **842** and a wire opening **858** of the wiring compartment **814**. As another example, an electrical cable that may be used in DC power connection may be routed through the wire opening **844** and a wire opening **860** of the wiring compartment **814**.

[0063] In some example embodiments, the sidewall **834** may include slots **850**, **852** and the sidewall **836** may include slots **854**, **856**. The slots **85-856** may be used to attach the power connection box **110** to the pole **104** of FIG. **1** using mounting bands such as the mounting band **116** shown in FIG. **1**.

[0064] In some alternative embodiments, the power connection box **110** may have a different shape than shown without departing from the scope of this disclosure. In some alternative embodiments, some elements of the power connection box **110**, such as openings, components, etc., may be at different locations than shown without departing from the scope of this disclosure. In some alternative embodiments, the AC compartment **802** and the DC compartment **804** may be switched without departing from the scope of this disclosure. In some alternative embodiments, the AC wiring section **862** and the DC wiring section **816** may be switched without departing from the scope of this disclosure. In some alternative embodiments, some elements of the power connection box **110** may be omitted without departing from the scope of this disclosure. For example, the AC wiring section **862** and the DC wiring section **816** may be omitted without departing from the scope of this disclosure.

[0065] FIG. **9** illustrates a remote power system **900** of an outdoor lighting system according to another example embodiment. In some example embodiments, the remote power system **900** includes remote power unit assemblies **904**, **906**, **908**, **910** and a power connection box **912** that are attached to a pole **902**. For example, the remote power unit assemblies **904** and **908** may be attached to the pole **902** using mounting structures such as a mounting fastener **914**, and the remote power unit assemblies **906** and **910** may be attached to the pole **902** using structures such as a mounting fastener **916** as can be readily understood by those of ordinary skill in the art. The power connection box **912** may be attached to the pole **902** using structures such as the mounting fastener **918** as can be readily understood by those of ordinary skill in the art.

[0066] In some example embodiments, the remote power system **900** may provide DC power to lightheads (e.g., LED lightheads) that are at a top end portion of the pole **902** in the manner described with respect to the remote power system **102** of FIG. **2**. For example, the remote power system **900** may provide DC power to a lighthead assembly such as the lighthead assembly **120** shown in FIG. **1**.

[0067] In some example embodiments, electrical cables between the remote power unit assemblies **904**, **906** and the power connection box **912** may be routed generally in the manner described with respect to the remote power system **102** of FIG. **2**. Because the remote power unit assemblies **908**, **910** are on opposite side of the pole **902** from the power connection box **912**, electrical cable between the remote power unit assemblies **908**, **910** and the power connection box **912** may be at least partially routed through flexible conduits **920** and **922** as can be readily understood by those of ordinary skill in the art. In general, electrical cables that are used for AC power connection between each one of the remote power unit assemblies **904-910** and power connection box **912** may be routed separated from electrical cables that are used for DC power connection between the respective one of the remote power unit assemblies **904-910** and power connection box **912**.

[0068] In some example embodiments, the power connection box **912** may include separate compartments for AC and DC components and connections as described below with respect to FIGS. **13**A and **13**B. For example, electrical connections used for AC power connection may be routed to one of the compartments of the power connection box **912**, and electrical connections used for DC power connection may be routed to another compartment of the power connection box **912**. One of the compartments of the power connection box **912** may be accessed by opening a door **924** of the power connection box **912**, and the other compartment may be access by opening another door that is on the opposite side of the power connection box **912** from the door **924**.

[0069] Because the remote power system **900** can provide DC power to a lighthead assembly while remotely located from the lighthead assembly, lightheads of the lighthead assembly **120** do not need to include local power supplies. By eliminating the need for lightheads to incorporate a driver, the remote power system **900** enables replacement of non-LED lightheads with LED lightheads.

[0070] In some alternative embodiments, the remote power system **900** may include more or fewer than four remote power unit assemblies shown in FIG. **9** without departing from the scope of this disclosure. In some alternative embodiments, one or more of the power unit assemblies **904-910** and the power connection box **912** may be attached to the pole **902** at different locations than shown without departing from the scope of this disclosure.

[0071] FIG. **10** illustrates a remote power unit assembly **1000** for use in the remote power system **900** of FIG. **9** according to an example embodiment. The remote power unit assembly **1000** corresponds to each one of the remote power unit assemblies **904-910** of the remote power system **900** of FIG. **9**. Referring to FIGS. **9** and **10**, in some example embodiments, the remote power unit assembly **1000** may include a mounting rack **1002** and remote power units **1004**, **1006**, **1008** attached to the mounting rack **1002**. For example, the remote power unit **1004** may be attached to the mounting rack **1002** using at least fasteners **1010** (e.g., screws). The remote power unit **1006** may be attached to the mounting rack **1002** using at least fasteners **1012** (e.g., screws). The remote power unit **1008** may be attached to the mounting rack **1002** using at least fasteners **1014** (e.g., screws).

[0072] In some example embodiments, each one of the remote power units **1004**, **1006**, **1008** may receive AC power and may generate DC power from the AC power. For example, each one of the remote power units **1004**, **1006**, **1008** may include one or more drivers (e.g., LED drivers) that generate DC power compatible with LED lightheads. The remote power units **1004**, **1006**, **1008** may each be connected to the power connection box **912** that receives AC power over one or more electrical cables that are routed through the pole **902**. The remote power units **1004**, **1006**, **1008** may be electrically coupled to the power connection box **912** and may receive AC power over an electrical cable **1016**.

[0073] In some example embodiments, the remote power units **1004**, **1006**, **1008** may be electrically connected to the power connection box **912** via an electrical cable **1018** to provide DC power to lightheads at a top end portion of the pole **902** through the power connection box **912**. To illustrate, DC power from the remote power units **1004**, **1006**, **1008** may be provided to lightheads over one or more electrical cables that are routed from the power connection box **912** through the inside of the pole **902**.

[0074] In some example embodiments, the electrical cables **1016**, **1018** may be routed through respective wiring conduits (e.g., wiring conduits **1116**, **1118** shown in FIGS. **11**A-**11**C) of the mounting rack **1002** and may each be terminated at a respective connector (not shown) at a bottom end of the remote power unit assembly **1000**. One or more wires of each of the electrical cables **1016**, **1018** may also be terminated at one or more connectors **1020** at a top end of the remote power unit assembly **1000**. One or more wires of each of the electrical cables **1016**, **1018** may also be terminated inside wiring conduits (e.g., wiring conduits **1116**, **1118** shown in FIGS. **11**A-**11**C) of the mounting rack **1002** and electrically connected to a respective one of the remote power units **1004**, **1006**, **1008**.

[0075] In some alternative embodiments, the remote power unit assembly **1000** may include more or fewer than three remote power units without departing from the scope of this disclosure. In some alternative embodiments, the electrical cables **1016**, **1018** may be terminated inside the mounting rack **1002** at a bottom end of the remote power unit assembly **1000** without departing from the scope of this disclosure. In some alternative embodiments, the remote power units **1004**, **1006**, **1008** may be attached to the mounting rack **1002** using means other than the fasteners **1010**, **1012**, **1014** without departing from the scope of this disclosure.

[0076] FIGS. **11**A-**11**C illustrate different views of the mounting rack **1002** of the remote power unit assembly **1000** of FIG. **10** used for attaching the remote power unit assembly of FIG. **10** to the pole **902** of FIG. **9** according to an example embodiment. Referring to FIGS. **11**A-**11**C, in some example embodiments, the mounting rack **1002** may include side frames **1102**, **1104**, a top cover **1106**, and a bottom cover **1108**. The mounting rack **1002** may also include driver shelves **110**, **112**, **114** that are securing attached to the side frames **1102**, **1104**. For example, the driver shelves **110**, **112**, **114** may have tabs that are inserted in respective slots of the side frames **1102**, **1104**. Alternatively or in addition, the driver shelves **110**, **112**, **114** may be attached to the side frames **1102**, **1104** using fasteners, welding, brackets, and/or other means as can be readily understood by those of ordinary skill in the art. The top cover **1106** and the bottom cover **1108** may also be attached to the side frames **1102**, 1104 using fasteners, welding, brackets, and/or other means as can be readily understood by those of ordinary skill in the art.

[0077] In some example embodiments, the mounting rack **1002** may include wiring conduits **1116**, **1118**. For example, the wiring conduits **1116**, **1118** may be separate structures or parts of an integral unit and may be attached to the side frames **1102**, **1104**. For example, the wiring conduits **1116**, **1118** may be attached to the side frames **1102**, **1104** using fasteners, welding, brackets, and/or other means as can be readily understood by those of ordinary skill in the art.

[0078] In some example embodiments, each one of the driver shelves **1110**, **1112**, **1114** is designed to hold a remote power unit. To illustrate, the remote power unit **1004** is positioned on the driver shelf **1110**, the remote power unit **1006** is positioned on the driver shelf **1112**, and the remote power unit **1008** is positioned on the driver shelf **1114**. The driver shelves **1110**, **1112**, **1114** may each include guides, such as a guide **1130** of the driver shelf **1112**, to facilitate the placement of the remote power units **1004**, **1006**, **1008** on a respective one of the driver shelves **1110**, **1112**, **1114**. The driver shelves **1110**, **1112**, **1114** may each include multiple openings, such as an opening **1132** of the driver shelf **1112**, for example, to reduce the overall weight of the mounting rack **1002**.

[0079] In some example embodiments, the top cover **1106** may include vents **1126** for dissipating heat generated by the remote power units **1004**, **1006**, **1008**. The bottom cover **1108** may also include vents **1128** for dissipating heat generated by the remote power units **1004**, **1006**, **1008**.

[0080] In some example embodiments, a removable panel **1122** may be attached to the side frame **1102** using, for example, one or more fasteners. A removable panel **1124** may be attached to the side frame **1104** using, for example, one or more fasteners such as a fastener (e.g., a screw). The removable panels **1122**, **1124** may include vents for dissipating heat generated by the remote power units **1004**, **1006**, **1008**. For example, the removable panel **1124** may include vents such as the vent **1138**.

[0081] In some example embodiments, the side frame **1102** may include a flange **1134** that may help retain the wiring conduit **1116** attached to the side frame **1102**. The side frame **1104** may include a flange **1136** that may help retain the wiring conduit **1118** attached to the side frame **1104**. For example, the flange **1134** may be welded to the wiring conduit **1116**, and the flange **1136** may be welded to the wiring conduit **1118**. The wiring conduit **1116** may have openings **1144** and **1140** for routing electrical cables in, out, and/or through the wiring conduit **1116**. For example, the electrical cable **1016** may be used for AC power connection, and the wiring conduit **1116** may be used to route the electrical cable **1016** (shown in FIG. **10**) therethrough. The wiring conduit **1118** may have openings **1146** and **1142** for routing electrical cables in, out, and/or through the wiring conduit **1118**. For example, the electrical cable **1018** may be used for DC power connection, and the wiring conduit **1118** may be used to route the electrical cable **1018** (shown in FIG. **10**) therethrough. By providing separate pathways for the electrical cables **1016**, **1018**, the wiring conduits **1116**, **1118** may enable the separate routing of AC power electrical cables from DC power electrical cables.

[0082] In some example embodiments, the wiring conduit **1116** may include a removable door **1120** that provides access to a cavity of the wiring conduit **1116** as more clearly shown in FIG. **11**C. For example, wire connections and adjustments in a cavity of the wiring conduit **1116** may be made by removing the removable door **1120** and accessing the cavity. To illustrate, the removable door **1120** may be removed, for example, by removing one or more fasteners that attach the removable door **1120** to the wiring conduit **1116**. In some example embodiments, the wiring conduit **1118** may also include a removable door similar to the removable door **1120**.

[0083] In some example embodiments, the mounting rack **1002** may be made from sheet metal, such as aluminum and/or steel sheet metal. The mounting rack **1002** may be made using methods such as cutting, bending, welding, fastening, etc.

[0084] In some alternative embodiments, the mounting rack **1002** may have more or fewer than three driver shelves without departing from the scope of this disclosure. In some alternative embodiments, one or more components of the mounting rack **1002** may have a different shape than shown without departing from the scope of this disclosure. For example, the wiring conduits **1116**, **1118** may be a round or rectangular cross section for the entire lengths. In some alternative embodiments, the driver shelves **1110**, **1112**, **1114** may have a different shape and/or more or fewer openings (e.g., the opening **1132** of the driver shelf **1112**) than shown without departing from the scope of this disclosure. In some alternative embodiments, one or more guides (e.g., the guide **1130** of the driver shelf **1112**) of the driver shelves **1110**, **1112**, **1114** may be omitted without departing from the scope of this disclosure.

[0085] FIGS. **12**A and **12**B illustrate the mounting rack **1002** of FIGS. **11**A-**4**C with the removable panels **1122**, **1124** removed for clarity of illustration according to an example embodiment. In some example embodiments, electrical cables **1202**, **1204**, **1206** are coupled to respective connectors attached to a sidewall **1220** of the wiring conduit **1118** as more clearly shown in FIG. **12**A. For example, the electrical cable **1202** may include connectors **1208** and **1210** at opposite ends of the electrical cable **1202**. The connector **1208** is designed to be connected to the remote power unit **1004** shown in FIG. **10**, and the connector **1210** may be connected to a connector **1212** that is attached to the sidewall **1220**. The electrical cable **1204** may be electrically connected to the remote power unit **1006**, and the electrical cable **1206** may be electrically connected to the remote power unit **1008** shown in FIG. **10**.

[0086] In some example embodiments, the electrical cable **1202** may be electrically connected to one or more wires of the electrical cable **1018** via the connectors **1210**, **1212**, establishing an electrical connection between the remote power unit **1004** (when the connector **1208** is attached to the remote power unit **1004**) and the electrical cable **1018**. For example, the electrical cable **1202** may be used to provide DC power from the remote power unit **1004**. The electrical cables **1204** and **1206** may also be electrically connected to respective one or more wires of the electrical cable **1018** and may be used to provide DC power from the respective one of the remote power units **1006**, **1008** shown in FIG. **10**.

[0087] In some example embodiments, electrical cables **1222**, **1224**, **1226** are coupled to respective connectors attached to a sidewall **1234** of the wiring conduit **1116** as more clearly shown in FIG. **12**B. For example, the electrical cable **1222** may include connectors **1228** and **1230** at opposite ends of the electrical cable **1222**. The connector

1228 is designed to be connected to the remote power unit **1004** shown in FIG. **10**, and the connector **1230** may be connected to a connector **1232** that is attached to the sidewall **1234**. The electrical cable **1224** may be electrically connected to the remote power unit **1006**, and the electrical cable **1226** may be electrically connected to the remote power unit **1008** shown in FIG. **10**.

[0088] In some example embodiments, the electrical cable **1222** may be electrically connected to one or more wires of the electrical cable **1016** via the connectors **1230**, **1232**, establishing an electrical connection between the remote power unit **1004** (when the connector **1228** is attached to the remote power unit **1004**) and the electrical cable **1016**. For example, the electrical cable **1222** may be used to provide AC power to the remote power unit **1004**. The electrical cables **1224** and **1226** may also be electrically connected to respective one or more wires of the electrical cable **1016** and may be used to provide AC power to the respective one of the remote power units **1006**, **1008** shown in FIG. **10**.

[0089] In some example embodiments, the mounting rack **1002** may include brackets **1214** and **1216** as more clearly shown in FIG. **12**B. For example, the bracket **1214** may be attached to the top cover **1106** by fasteners and/or other means such as welding. The bracket **1214** may include tabs **1218** that may be used to retain a mounting rod that can be used in mounting the mounting rack **1004** attached to the pole **902** shown in FIG. **9**. The bracket **1216** may be attached (e.g., welded) to the wiring conduits **1116**, **1118** and may be used in mounting the mounting rack **1002** to the pole **902**. For example, the bracket **1216** may include a tab **1236** that may be used to retain a mounting rod that can be used in mounting the mounting rack **1004** to the pole **902**.

[0090] By providing separate wiring conduits for AC and DC power connections, the mounting rack **1002** improves the separation of AC and DC power, which can, for example, reduce signal interference. By using the electrical cables **1202-1206**, **1222-1226** that are connected to either the electrical cable **1016** or **1018** using connectors that are wired inside the wiring conduits **1116**, **1118**, the remote power units **1004**, **1006**, **1008** may be conveniently connected to the electrical cables **1016**, **1018** to establish electrical connections with the power connection box **912** shown in FIG. **9**.

[0091] In some alternative embodiments, one or more connectors of the electrical cables **1202-1206**, **1222-1226** may be coupled at different sides of the wiring conduits **1116**, **1118** than shown without departing from the scope of this disclosure. In some alternative embodiments, one or more connectors of the electrical cables **1202-1206**, **1222-1226** may be omitted without departing from the scope of this disclosure.

[0092] FIGS. **13**A and **13**B illustrate different views of the power connection box **912** of the remote power system **900** of FIG. **9** according to an example embodiment. In FIGS. **13**A and **13**B, the power connection box **912** is shown without doors, such as the door **924** shown in FIG. **9**, for clarity of illustration. The power connection box **912** may be made from sheet metal, such as aluminum and/or steel sheet metal, using methods such as cutting, bending, welding, etc.

[0093] In some example embodiments, the power connection box **912** may include an AC compartment **1302** and a DC compartment **1322** that are separated from each other by a backwall **1304**. The AC compartment **1302** may contain components such as a surge protector **1306**, circuit breakers **1308**, etc. that are attached to a back plate **1310**. The back plate **1310** may be attached to the backwall **1304** by one or more fasteners, such as the fastener **1320**.

[0094] In some example embodiments, the DC compartment **802** may include components, such as wire connectors **1324**, etc., that are attached to a back plate **1326**. The back plate **1326** may be attached to the backwall **1304** using one or more fasteners, such as the fastener **1328**. The power connection box **110** may be made from sheet metal, such as aluminum and/or steel sheet metal, using methods such as cutting, bending, welding, etc.

[0095] In some example embodiments, the power connection box **912** may include wire conduits **1312**, **1314** for routing sections of electrical cables that are routed between the power connection box **912** and, for example, the remote power unit assemblies **904**, **906** shown in FIG. **9**. For example, the wire conduit **1312** may be used to route electrical cables that are used for AC power connection, and the wire conduit **1314** may be used to route electrical cables that are used for DC power connection. To illustrate, the electrical cable **1016** or another electrical cable coupled to the electrical cable **1016** may be routed through the wire conduit **1312** and may be connected to one or more components in the AC compartment **1302** of the power connection box **912**. The electrical cable **1018** or another electrical cable coupled to the electrical cable **1018** may be routed through the wire conduit **1314** and may be connected to one or more components in the DC compartment **1322** of the power connection box **912**. In general, the wire conduits **1312** and **1314** are separated from each other and provide separated routed of AC power connection electrical cables from DC power connection electrical cables.

[0096] In some example embodiments, the power connection box **912** may also include wire conduits **1330**, **1332** for routing sections of electrical cables that are routed between the power connection box **912** and, for example, the remote power unit assemblies **908**, **910** of FIG. **9** through flexible conduits **920**, **922** shown in FIG. **9**. To illustrate, the flexible conduits **920** and **922** may be connected to the wire conduits **1330** and **1332**, respectively. For example, electrical cables that are used for AC power connection may be routed through the wire conduit **1330**, and electrical cables that are used for DC power connection may be routed through the wire conduit **1332**, or vice versa. In general, the wire conduits **1330** and **1332** are separated from each other and provide separated routed of AC power connection electrical cables from DC power connection electrical cables.

[0097] In some example embodiments, the power connection box **912** includes a wire opening **1316** that may be used for electrical wire routing between the power connection box **912** and the inside of the pole **902** shown in FIG. **9**. For example, the wire opening **1316** may be aligned with an opening in the pole **902** as can be readily understood by those of ordinary skill in the art.

[0098] By providing separate routing and compartments for AC power connections and DC power connections, the power connection box **912** can simplify reduce interference between AC power and DC power. The power connection box **912** can also simplify maintenance and repair by separating AC and DC components.

[0099] In some alternative embodiments, the power connection box **912** may have a different shape than shown without departing from the scope of this disclosure. In some alternative embodiments, some elements of the power connection box **912**, such as openings, components, etc., may be at different locations than shown without departing from the scope of this disclosure. In some alternative embodiments, the AC compartment **1302** and the DC compartment **1322** may be switched without departing from the scope of this disclosure. In some alternative embodiments, some elements of the power connection box **912** may be omitted without departing from the scope of this disclosure. For example, the wire conduits **1330**, **1332** may be omitted without departing from the scope of this disclosure. In some alternative embodiments, the power connection box **912** may include another pair of wire conduits that may be used for separate routing of AC and DC electrical cables without departing from the scope of this disclosure.

[0100] Although particular embodiments have been described herein in detail, the descriptions are by way of example. The features of the example embodiments described herein are representative and, in alternative embodiments, certain features, elements, and/or steps may be added or omitted. Additionally, modifications to aspects of the example embodiments described herein may be made by those skilled in the art without departing from the scope of the following claims, the scope of which are to be accorded the broadest interpretation so as to encompass modifications and equivalent structures.

1. A mounting rack for attachment of a remote power unit to a structure, the mounting rack comprising:
- a first side frame;
- a second side frame;
- a separator plate, wherein the first side frame and the separator plate provide an alternating current (AC) wiring pathway therebetween for routing a first electrical cable, wherein the second side frame and the separator plate provide a direct current (DC) wiring pathway therebetween for routing a second electrical cable, and wherein the AC wiring pathway and the DC wiring pathway are separated by the separator plate;
- a bottom plate attached to the first side frame and the second side frame, the bottom plate having a first wire opening aligned with the AC wiring pathway and a second wire opening aligned with the DC wiring pathway; and
- a top plate attached to the first side frame and the second side frame, the top plate having a third wire opening aligned with the AC wiring pathway and a fourth wire opening aligned with the DC wiring pathway.

2. The mounting rack of claim **1**, further comprising a front plate having a third wire opening aligned with the AC wiring pathway and a fourth wire opening aligned with the DC wiring pathway.

3. The mounting rack of claim **1**, wherein the first side frame and the second side frame each comprise a fastener hole for attaching the remote power unit to the mounting rack using one or more fasteners.

4. The mounting rack of claim **3**, further comprising a hollow tube aligned with the first fastener hole and the second fastener hole, wherein the hollow tube extends through a hole in the separator plate.

5. The mounting rack of claim **3**, wherein the first side frame and the second side frame each further comprise a second fastener hole for attaching the remote power unit to the mounting rack using second one or more fasteners.

6. The mounting rack of claim **1**, wherein the first side frame and the second side frame each comprise one or more slots for extending one or more mounting bands therethrough for attaching the mounting rack to the structure.

7. The mounting rack of claim **1**, further comprising a support rung extending between and attached to the first side frame and the second side frame.

8. A remote power unit assembly, comprising:
- a remote power unit;
- a bracket attached to the remote power unit; and
- a mounting rack, comprising:
- a first side frame;
- a second side frame; and
- a separator plate, wherein the first side frame and the separator plate provide an alternating current (AC) wiring pathway therebetween for routing a first electrical cable, wherein the second side frame and the separator plate provide a direct current (DC) wiring pathway therebetween for routing a second electrical cable, wherein the AC wiring pathway and the DC wiring pathway are separated by the separator plate, and wherein the bracket is attached to the first side frame and the second side frame;
- wherein the first side frame and the second side frame each comprise a fastener hole, and wherein the bracket is attached to the first side frame and the second side frame using one or more fasteners extending through the fastener hole of each of the first side frame and the second side frame; and
- wherein the mounting rack further comprises a hollow tube aligned with the fastener hole of each of the first side frame and the second side frame and wherein the hollow tube extends through a hole in the separator plate.

9. The remote power unit assembly of claim **8**, wherein the mounting rack further comprises a bottom plate attached to the first side frame and the second side frame, the bottom plate having a first wire opening aligned with the AC wiring pathway and a second wire opening aligned with the DC wiring pathway.

10. The remote power unit assembly of claim **9**, wherein the mounting rack further comprises a front plate having a third wire opening aligned with the AC wiring pathway and a fourth wire opening aligned with the DC wiring pathway such that the first electrical cable is routed from the remote power unit through the third wire opening, a portion of the AC wiring pathway, and the first wire opening and such that the second electrical cable is routed from the remote power unit through the fourth wire opening, a portion of the DC wiring pathway, and the second wire opening.

11. The remote power unit assembly of claim **8**, wherein the first side frame and the second side frame each comprise one or more slots for extending one or more mounting bands therethrough for attaching the mounting rack to a structure.

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